

Journal Pre-proof

Joint scheduling and resource allocation based on reinforcement learning
in integrated access and backhaul networks

Joeun Kim, Youngil Jeon, Junhwan Lee, Moon-Sik Lee, Taesoo Kwon



PII: S2405-9595(25)00032-3
DOI: <https://doi.org/10.1016/j.icte.2025.03.004>
Reference: ICTE 823

To appear in: *ICT Express*

Received date : 11 December 2024
Revised date : 7 February 2025
Accepted date : 5 March 2025

Please cite this article as: J. Kim, Y. Jeon, J. Lee et al., Joint scheduling and resource allocation based on reinforcement learning in integrated access and backhaul networks, *ICT Express* (2025), doi: <https://doi.org/10.1016/j.icte.2025.03.004>.

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2025 Published by Elsevier B.V. on behalf of The Korean Institute of Communications and Information Sciences. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Joint Scheduling and Resource Allocation Based on Reinforcement Learning in Integrated Access and Backhaul Networks

Jooun Kim^a, Youngil Jeon^b, Junhwan Lee^b, Moon-Sik Lee^b, Taesoo Kwon^{a,*}

^aSeoul National University of Science and Technology, Seoul, South Korea

^bElectronics and Telecommunications Research Institute, Daejeon, South Korea

Abstract

In recent wireless networks, the integrated access and backhaul (IAB) network provides a cost-effective solution to enhance network performance and has emerged as a key technology not only for beyond 5G but also for 6G. Because of the inherent nature of IAB, access and backhaul links share the same resource pool causing cross-link interference. To address this challenge, this paper investigates an algorithm based on reinforcement learning (RL) for joint scheduling and resource allocation (RA) problem, aiming to mitigate interference and enhance user data rates. However, the scale of this joint problem is too large to solve using RL alone. Therefore, this paper proposes decomposing the joint problem into virtual scheduling and RL-based RA (RL-RA), and then solving them collaboratively. Simulation results also show that the proposed algorithm significantly improves performance and can be applied comprehensively to various duplex modes, including half and full duplex types, and different frequency bands, such as sub-6GHz and mmWave.

© 2024 The Author(s). Published by Elsevier B. V. on behalf of The Korean Institute of Communications and Information Sciences. This is an open access article under the CCBY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

Keywords: Reinforcement learning, integrated access and backhaul, B5G/6G, resource allocation, scheduling

1. Introduction

More aggressive base station (BS) densification and frequency reuse strategies can dramatically improve user experienced data rates. However, installing numerous BSs is realistically challenging due to the high installation cost. Therefore, the integrated access and backhaul (IAB) network, which offers a simple solution for densification, is a key technology for B5G, and its concept is expected to be extended and applied in 6G. [1, 2]. IAB comprises a donor BS (DBS) connected to a core network via a wired backhaul and IAB nodes provided wireless backhaul by DBS. While IAB nodes can be readily installed using wireless backhaul, a wireless backhaul link shares the same resource pool as an access link, necessitating consideration of cross-link interference and requiring meticulous resource allocation (RA) [3, 4]. In this regard, this paper investigates algorithms for joint scheduling and RA to enhance IAB performance.

Numerous studies have addressed RA problem in IAB networks, such as those in [5]–[10]. Several works [5, 6] empirically found the optimal bandwidth partition between access and backhaul that maximizes rate coverage under stochastic geometry frameworks. However, these approaches do not precisely reflect dynamic environment changes, such as variations

in mutual interference by user movement. Recent literature frequently employs various mathematical optimization techniques to solve the RA problem [11]–[13], while RL-based approaches are also widely used [14]. Moreover, RL is extensively utilized to effectively solve various problems in B5G networks. [15, 16] This integration of artificial intelligence and communication represents one of the key scenarios in 6G and can be a crucial part of its development [17]. Accordingly, recent literature has explored the use of reinforcement learning (RL) for greater adaptability to the dynamic IAB environment. For instance, RL was used to address scheduling or various RA problems in IAB, including spectrum allocation [7, 8], power allocation [9], their joint optimization [10], and scheduling [18]. While [7, 9] did not support multiuser scenario, [10] did. However, [10] allocated only one of the multiple subchannels to each link for simplicity. Meanwhile, RL for RA can be found more in non-IAB, e.g., in cellular [19], mmWave nonorthogonal multiple access [20], unmanned aerial vehicle [21] networks, and hierarchical federated learning system [22]. [22] addressed the RA and client scheduling problem in two phases. In the first phase, it optimized client selection, power control, and CPU configuration based on RL. However, this stage did not consider orthogonal frequency division multiplexing, and this study was conducted in a static environment where the clients did not move.

The joint scheduling and RA problem in IAB is expressed as NP-hard integer programming and requires a large amount of runtime and resources, thus its scale is too large to solve by RL alone. To the best of our knowledge, no study has addressed

*Corresponding author

Email addresses: worth3910@seoultech.ac.kr (Jooun Kim), youngil@etri.re.kr (Youngil Jeon), junhwanlee@etri.re.kr (Junhwan Lee), moonsiklee@etri.re.kr (Moon-Sik Lee), tskwon@seoultech.ac.kr (Taesoo Kwon)

this joint problem in IAB using RL.

Therefore, this paper proposes an algorithm that decomposes the joint scheduling and RA problem into the two subproblems, i.e., *virtual scheduling* and RL-based RA (referred to as *RL-RA*), which collaboratively work. The decomposition approach is often used for large-scale problems, and decomposed subproblems' collaborative interaction is elaborately designed to yield even better performance [23]. The novelties of this paper are highlighted as follows:

- (i) The joint problem is decomposed into virtual scheduling and RL-RA, whose collaborative operation improves performance. First, virtual scheduling reduces the multiuser complexity by choosing users to be potentially served. Following this, RL-RA achieves proportional fair RA for the IAB network by coordinating RA across backhaul and access links only for potentially scheduled users. In the RL design, state information is significantly reduced to enhance efficiency.
- (ii) This paper also demonstrates that the proposed algorithm can be applied comprehensively to various duplexing (i.e., half and full duplex) and frequency bands (e.g., sub-6GHz and mmWave). Although the mmWave is more complex than the sub-6GHz because of prominent features such as beamforming and blockage effects, the proposed algorithm still performs well.

Notations: $\mathbb{E}[X]$ and superscript T denote the expectation of random variable X and the transpose of a matrix, respectively. And $\mathbb{B}^{M \times N}$ denote the set of binary matrices with dimension $M \times N$ consisting of 0 and 1.

2. IAB Network Models

This paper considers the IAB network working over either sub-6GHz or mmWave band. The total bandwidth is divided into M orthogonal subchannels (subchs) and is denoted by $\mathcal{M} \triangleq \{1, \dots, M\}$. There is a single DBS and it serves L IAB nodes with wireless backhaul links. BS includes both DBS and all IAB nodes, and the set of BSs is denoted as $\mathcal{L} \triangleq \{0, 1, 2, \dots, L\}$, where BS 0 represents the DBS index. Hence, the set of IAB nodes becomes $\mathcal{L}^- \triangleq \mathcal{L} \setminus \{0\}$. Users are deployed with the homogeneous Poisson point process (HPPP). A user is associated with the BS providing the strongest average received power, and a BS can serve multiple users. Here, when K_l users are associated to BS l , their set is denoted by $\mathcal{K}_l \triangleq \{(k, l) | k = 1, \dots, K_l\}$. BS l chooses the users to whom subchs are allocated among these K_l users. Fig. 1 illustrates this system model.

The received power of receiver j from transmitter i on subch $m \in \mathcal{M}$, $g_{i,j}^m$, can be expressed as $g_{i,j}^m = (P_i/M)h_{i,j}^m L(D_{i,j})\varphi_{i,j}^m$, where P_i is the total transmit power of transmitter i and $h_{i,j}^m$ is the small-scale fading gain modeled by Nakagami- m fading. $h_{i,j}^m$ is independent among the different subchs. The large-scale fading gain $L(D_{i,j})$ is $\beta D_{i,j}^{-\alpha}$, where β is the path loss gain at one meter, $D_{i,j}$ is the distance between transmitter i and receiver j ,

and α is the path loss exponent. $\varphi_{i,j}^m$ is the antenna gain between i and j , and the BSs and users have a single antenna in sub-6GHz. On the other hand, the mmWave exhibits more unique channel features than sub-6GHz. The mmWave channel is sensitive to blockages, so the channel is classified into line-of-sight (LoS) and non-line-of-sight (NLoS). The channel becomes LoS with a probability of $e^{-D_{i,j}/\mu}$, and NLoS with a probability of $1 - e^{-D_{i,j}/\mu}$, where μ is the LoS range constant [24]. Consequently, if a channel is LoS, α is α_L ; otherwise, α_N . Additionally, m of Nakagami- m fading depends on LoS and NLoS. Beamforming is another key feature of mmWave, capable of compensating for harsh path loss. The antenna gain is approximated by a sectorized beamforming model [24], where a receiver obtains the main lobe gain within the beamwidth and the side lobe gain otherwise.

3. Problem Formulation for Joint Scheduling and Resource Allocation

This section formulates the joint scheduling and RA problem. The scheduling and RA vectors are defined as the $M \times 1$ column vectors of which each element is either 0 or 1 as follows: (i) the vector for links from the DBS to its associated users or IAB nodes (i.e., *DBS links*) is $\mathbf{y}_j \triangleq [y_j^1 y_j^2 \dots y_j^M]^T$ for $j \in \mathcal{K}_0 \cup \mathcal{L}^-$; (ii) the vector for links from an IAB node to its associated users (i.e., *IAB node links*) is $\mathbf{z}_k \triangleq [z_k^1 z_k^2 \dots z_k^M]^T$, where $k \in \mathcal{K}_l$ for a given $l \in \mathcal{L}^-$. If subch m is allocated to receiver j or k , the corresponding y_j^m or z_k^m is set to 1, and otherwise 0. Therefore, the scheduling and RA matrices are defined as $\mathbf{Y} \triangleq [\mathbf{y}_{(1,0)} \dots \mathbf{y}_{(K_0,0)} \mathbf{y}_1 \dots \mathbf{y}_L] \in \mathbb{B}^{M \times (K_0+L)}$ for DBS links and $\mathbf{Z} \triangleq [\mathbf{z}_{(1,1)} \dots \mathbf{z}_{(K_1,1)} \dots \mathbf{z}_{(K_L,L)}] \in \mathbb{B}^{M \times \sum_{l=1}^L K_l}$ for IAB nodes links. Hence, y_j^m and z_k^m represent the (m, j) th element of $\mathbf{Y}$ and the (m, k) th element of $\mathbf{Z}$, respectively.

The signal to interference plus noise ratio (SINR) of the DBS link for user $k_0 \in \mathcal{K}_0$ is expressed as $\text{SINR}_{k_0}^m \triangleq g_{0,k_0}^m y_{k_0}^m / (\sum_{l \in \mathcal{L}^-} (g_{l,k_0}^m \sum_{k \in \mathcal{K}_l} z_k^m) + \sigma^2)$, where σ^2 is the noise power. The SINR of a backhaul link for IAB node $l \in \mathcal{L}^-$ is $\text{SINR}_l^m \triangleq g_{0,l}^m y_l^m / (\sum_{i \in \mathcal{L}^- \setminus \{l\}} (g_{i,l}^m \sum_{k \in \mathcal{K}_i} z_k^m) + V_l^m + \sigma^2)$, where V_l^m is the self-interference and expressed as $(P_l/M)\xi \sum_{k \in \mathcal{K}_l} z_k^m$. ξ denotes the self-interference cancellation capability in the formula. The system supports either of full duplex or half duplex. The full duplex is modeled with residual self-interference, while the half duplex has a constraint such that z_k^m equals 0 when y_l^m equals 1 for $k \in \mathcal{K}_l$. The SINR of the IAB node link for user $k_l \in \mathcal{K}_l$ is $\text{SINR}_{k_l}^m \triangleq g_{l,k_l}^m z_{k_l}^m / (\sum_{i \in \mathcal{L}^- \setminus \{l\}} (g_{i,k_l}^m \sum_{k \in \mathcal{K}_i} z_k^m) + g_{0,k_l}^m \sum_{j \in \mathcal{K}_0 \cup \mathcal{L}^-} y_j^m + \sigma^2)$. As a result, the achievable rate of user $k_0 \in \mathcal{K}_0$ served by the DBS is expressed as $R_{k_0} \triangleq \sum_{m \in \mathcal{M}} \frac{W}{M} \log(1 + \text{SINR}_{k_0}^m)$. Also, for user $k_l \in \mathcal{K}_l$ served by IAB node l for $l \in \mathcal{L}^-$, the access rate is given by $r_{k_l} \triangleq \sum_{m \in \mathcal{M}} \frac{W}{M} \log(1 + \text{SINR}_{k_l}^m)$, and the achievable rate becomes the minimum value of the backhaul and access rates, i.e., $R_{k_l} \triangleq \min\{\eta_{k_l} \sum_{m \in \mathcal{M}} \frac{W}{M} \log(1 + \text{SINR}_l^m), r_{k_l}\}$, where η_{k_l} denotes the ratio of the subchs for user k_l to the total subchs that IAB node l uses for its backhaul link. This paper simply models η_{k_l} as follows: (i) the *equal partition* equally allocates the backhaul subchs of IAB l across users, i.e.,

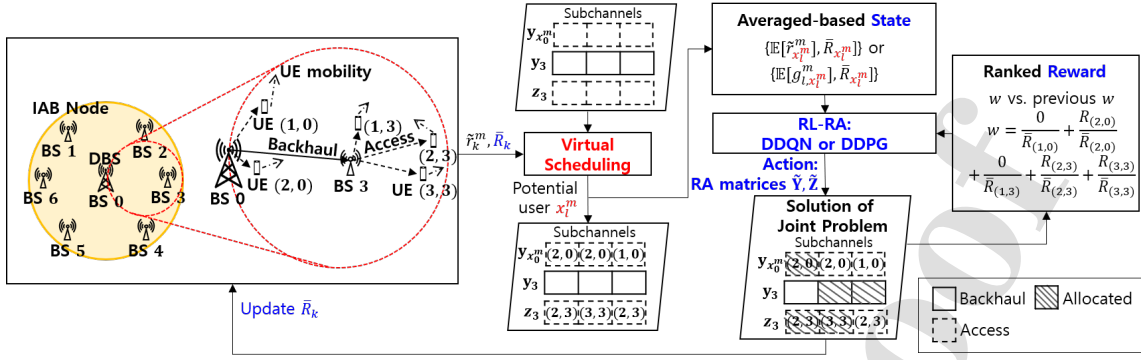


Figure 1. System models and collaborative operations for virtual scheduling and RL-RA.

$\eta_{k_l} \triangleq 1 / \sum_{k \in \mathcal{K}_l} \vee_{m \in \mathcal{M}} z_k^m$ where $\vee$ denotes a logical OR operator; or (ii) the *access-subch-based partition* allocates its subchs proportional to the amount of access link subchs assigned to its users, i.e., $\eta_{k_l} \triangleq \sum_{m \in \mathcal{M}} z_{k_l}^m / \sum_{m \in \mathcal{M}} \sum_{k \in \mathcal{K}_l} z_k^m$. The impact of η_{k_l} on the performances will be examined in Section 5.

The objective function of the joint scheduling and RA problem is designed to maximize the sum of the logarithm of an individual user's long term average data rate. The long term average rate $\bar{R}_k$ for user k is recursively computed by $\gamma \bar{R}_k + (1 - \gamma) R_k$ with $0 < \gamma < 1$, where γ is a parameter that reflects the degree of importance of instantaneous data. Hence, this problem can be formulated as follows:

$$\underset{\mathbf{Y} \in \mathbb{B}^{M \times (K_0 + L)}, \mathbf{Z} \in \mathbb{B}^{M \times \sum_{l=1}^L K_l}}{\text{maximize}} \sum_{l \in \mathcal{L}} \sum_{k \in \mathcal{K}_l} \log(\bar{R}_k), \quad (1a)$$

$$\text{subject to} \sum_{j \in \mathcal{K}_0 \cup \mathcal{L}^-} y_j^m \leq 1, \forall m \in \mathcal{M}, \quad (1b)$$

$$\sum_{k \in \mathcal{K}_l} z_k^m \leq 1, \forall m \in \mathcal{M}, \forall l \in \mathcal{L}^-, \quad (1c)$$

where y_j^m and z_k^m are the (m, j) th element of $\mathbf{Y}$ and the (m, k) th element of $\mathbf{Z}$, respectively. (1b) and (1c) indicate that a BS can allocate each subch to only one receiver.

4. Reinforcement Learning-based Algorithm

The scale of the joint scheduling and RA problem is too large to be solved by RL alone. Therefore, this joint problem is decomposed into two subproblems, i.e., virtual scheduling and RL-RA, solved collaboratively. The virtual scheduling selects one user per each subch to be potentially served. This selected user is referred to *potential user* because the RL-RA determines later whether a potential user is actually assigned the subch. The RL-RA considers only these potential users as candidates for RA, thereby significantly reducing multiuser complexity, while allocating subchs to enhance proportional fairness across all users. Then, $\bar{R}_k$ is updated and again utilized in the virtual scheduling at the next step. Through continuous updating and learning at each step, optimization problem (1) is ultimately solved. Fig. 1 elaborates on the decomposition of the joint problem into the two subproblems and their collaboration.

4.1. Virtual Scheduling

The virtual scheduling basically pursues proportional fairness as in (1a), and also aims to reduce the complexity by allowing a BS to select one user independently for each subch, as in [25]. Hence, for a given subch $m \in \mathcal{M}$, BS $l \in \mathcal{L}$ independently chooses potential user x_l^m , as follows:

$$x_l^m \triangleq \arg \max_{k \in \mathcal{K}_l} \frac{\tilde{r}_k^m}{\bar{R}_k}, \quad (2)$$

where $\tilde{r}_k^m$ is the instantaneous rate of the user $k \in \mathcal{K}_l$ for $\forall l \in \mathcal{L}$, i.e., $\tilde{r}_k^m \triangleq \frac{W}{M} \log(1 + (g_{l,k}^m / \sum_{i \in \mathcal{L} \setminus \{l\}} g_{i,k}^m + \sigma^2))$. It considers only access links to reduce complexity, because IAB nodes use common backhaul links for all its users and independently select their potential users. As a result, in matrices $\mathbf{Y}$ and $\mathbf{Z}$, the virtual scheduling sets y_k^m for all $k \in \mathcal{K}_0 \setminus \{x_0^m\}$ and z_k^m all $k \in \mathcal{K}_l \setminus \{x_l^m\}$ to zeros. $\tilde{\mathbf{Y}}$ and $\tilde{\mathbf{Z}}$ denotes these simpler scheduling and RA matrices, like

$$\tilde{\mathbf{Y}} \triangleq \mathbf{Y} \text{ such that } y_k^m = 0, \forall k \in \mathcal{K}_0 \setminus \{x_0^m\}, \forall m \in \mathcal{M}. \quad (3)$$

$$\tilde{\mathbf{Z}} \triangleq \mathbf{Z} \text{ such that } z_k^m = 0, \forall k \in \mathcal{K}_l \setminus \{x_l^m\}, \forall m \in \mathcal{M}, \forall l \in \mathcal{L}^-. \quad (4)$$

4.2. RL-based Resource Allocation

Unlike problem (1) with the original $\mathbf{Y}$ and $\mathbf{Z}$, the RA-RL now tries to solve the reduced problem with $\tilde{\mathbf{Y}}$ and $\tilde{\mathbf{Z}}$, which is formulated as follows:

$$\underset{\tilde{\mathbf{Y}} \in \mathbb{B}^{M \times (1+L)}, \tilde{\mathbf{Z}} \in \mathbb{B}^{M \times L}}{\text{maximize}} \sum_{l \in \mathcal{L}} \sum_{k \in \mathcal{K}_l} \log(\bar{R}_k), \quad (5a)$$

$$\text{subject to} \sum_{j \in \{x_0^m\} \cup \mathcal{L}^-} \tilde{y}_j^m \leq 1, \forall m \in \mathcal{M}, \quad (5b)$$

where $\tilde{y}_j^m$ and $\tilde{z}_k^m$ are the (m, j) th element of $\tilde{\mathbf{Y}}$ and the (m, k) th element of $\tilde{\mathbf{Z}}$, respectively, which meet the conditions in (3) and (4). Note that $\bar{R}_k$ is expressed using $\tilde{y}_j^m$'s and $\tilde{z}_k^m$'s in $\tilde{\mathbf{Y}}$ and $\tilde{\mathbf{Z}}$. (5b) indicates that the DBS can serve only one receiver on each subch. It is worth noting that problem (1) with the original $\mathbf{Y}$ and $\mathbf{Z}$ has $M \times (L + \sum_{l=0}^L K_l)$ variables while problem

(4) with the simpler $\tilde{\mathbf{Y}}$ and $\tilde{\mathbf{Z}}$ has $M \times (2L + 1)$ variables. Thus, when RL is used to solve the RA subproblem (5), it now handles significantly fewer variables.

To solve the RA problem, this paper uses deep deterministic policy gradient (DDPG) RL algorithm. The DDPG is an actor-critic-based algorithm that extends the traditional actor-critic framework with experience replay and target networks. In this framework, the actor-network selects actions based on a learned policy, and the critic network evaluates those actions by estimating their values [26]. The key RL elements, including state, action, and reward, are designed to achieve the objective of (5) and Algorithm 1 describes the proposed joint algorithm using DDPG, as follows.

Algorithm 1 Via DDPG, Joint Virtual Scheduling and RL-RA

```

Initialize IAB network environment and replay buffer  $B$ ;
2: Initialize critic net  $Q(s, a | \theta^Q)$ , actor net  $\pi(s | \theta^\pi)$ ,
   target critic and actor nets with weight  $\theta^{Q'} = \theta^Q, \theta^{\pi'} = \theta^\pi$ ;
   for  $ep = 1:\text{numEp}$  do
4:   Observe initial scheduling and state;
       for  $t = 1:\text{numStep}$  do
6:     Obtain action  $a_t$  via  $\epsilon$ -greedy from actor net;
       Observe reward  $r_t$  by (6);
8:     Select a potential user on each subch by (2);
       Observe next state  $s_{t+1}$ ;
10:    Store  $(s_t, a_t, r_t, s_{t+1})$  into replay buffer  $B$ ;
       if train net update then
12:        Sample mini-batch of size  $N$  from replay buffer  $B$ ;
        Set  $y_i = r_i + \gamma Q'(s_{i+1}, \pi'(s_{i+1} | \theta^{\pi'}) | \theta^{Q'})$ ;
14:        Update critic net by  $\arg \min_{\theta^Q} \sum_{i=1}^N (y_i - Q(s_i, a_i | \theta^Q))^2$ ;
        Update actor net using sampled policy gradient
         $\nabla_{\theta^\pi} J \approx \mathbb{E}[\nabla_a Q(s, a | \theta^Q)]_{|s=s_i, a=\pi(s_i)} \nabla_{\theta^\pi} \pi(s | \theta^\pi)_{|s=s_i}$ ;
16:       end if
       Update target nets by
        $\theta^{Q'} = \tau \theta^Q + (1 - \tau) \theta^{Q'}, \theta^{\pi'} = \tau \theta^\pi + (1 - \tau) \theta^{\pi'}$ ;
18:   end for
end for

```

1) *State*: The current state s_t may be straightforwardly designed as a pair of the current instantaneous data rate $\tilde{r}_{k,t}^m$ and the current long term average rate $\tilde{R}_{k,t}$. However, because this paper considers the multi-user environment, the state space grows with the number of users and it is excessively large. Therefore, only potential users selected by virtual scheduling are considered, i.e., $k = x_t^m$. Additionally, to further reduce the state space, this paper uses the averaged value across all subchs and Section 5 will demonstrate that its performance is similar to that with the original raw values. Therefore, the current state s_t is designed as $s_t = \{\mathbb{E}[\tilde{r}_{x_t^m, t}^m], \tilde{R}_{x_t^m, t}\}_{m \in \mathcal{M}, l \in \mathcal{L}}$.

2) *Action*: The current action is defined by the RA matrices of the current step as $a_t = [\tilde{\mathbf{Y}}_t, \tilde{\mathbf{Z}}_t]$. The action space size is $(1 + L)^M$ for $\tilde{\mathbf{Y}}_t$ and $2^{(L \times M)}$ for $\tilde{\mathbf{Z}}_t$.

3) *Reward*: Note that [7] computed the reward independently

Table 1
Parameter notation and definition.

Notation	Definition
$\mathcal{M}$	Set of orthogonal subchannels
$\mathcal{L}$	Set of BSs
$\mathcal{L}^-$	Set of IAB nodes
$\mathcal{K}_l$	Set of users are associated to BS l
$\mathbf{Y}, \mathbf{Z}$	Scheduling and RA matrices for DBS, IAB nodes link
$\tilde{\mathbf{Y}}, \tilde{\mathbf{Z}}$	Simpler scheduling and RA matrices for DBS, IAB nodes link in (3),(4)
R_k	Achievable rate of user k
$\bar{R}_k$	Long term average rate for user k
$\tilde{r}_k^m$	Instantaneous rate of the user k
x_t^m	Potential user

Table 2
System parameters for IAB networks.

Parameter	Sub-6GHz	mmWave
Carrier frequency	4GHz	28GHz
W : Total bandwidth	20MHz	1GHz
M : Number of subchannels	10	
P_l : Max transmit power of BS l	DBS:44dBm IAB:33dBm	DBS:50dBm IAB:20dBm
β : Path loss gain at 1m	-44.48dB	-70dB
α : Path loss exponent	3	LoS:2, NLoS:3.3
m : Nakagami- m fading	1	LoS:2, NLoS:3
μ : LoS range constant	-	141.4m
Transmit(Tx)&Receive(Rx) beams ([main & side lobe gains, beam width])	BS Tx & User Rx: (0dB, 0dB, 360°) IAB Rx: (10dB, 0dB, 20°)	BS Tx: (18dB, -2dB, 10°) User Rx: (0dB, 0dB, 360°)
σ^2 : Noise power	-174dBm/Hz + 10log ₁₀ W + 10dB	
ξ : Self-interf. cancel. capa.	-110dB [28]	

for each step, because an IAB node only serves the fixed single user. In contrast, in the multiuser scenarios, the scheduled users are only served while the remaining users are not; i.e., one has to consider the reward for unserved as well as served users based on all users' accumulated satisfaction. Hence, this paper defines $w \triangleq \sum_{l \in \mathcal{L}} \sum_{k \in \mathcal{K}_l} R_k / \bar{R}_k$, and designs the ranked reward [27] that obtains the current reward value through comparing w at the current step with past statistics, as follows:

$$r_t = \begin{cases} 1, & w > w_{66} \\ 0, & w_{33} < w < w_{66} \\ -1, & \text{otherwise} \end{cases}, \quad (6)$$

where w_p denotes the p th percentile of the dataset of w values from the beginning step to the step just before.

The parameters defined in Section 2-4 are listed in Table 1.

5. Performance Evaluation and Discussion

As depicted in Fig. 1, a single DBS is located at the network's origin, with 6 IAB nodes positioned 250 meters away from the DBS at equal angular positions. Additionally, 30 users are deployed within a 400 meter network according to HPPP. At every step, users move in a specific direction, which changes randomly upon reaching the simulation area's boundary; the user

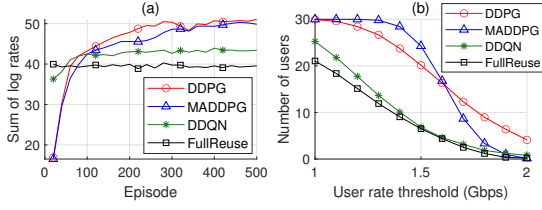


Figure 2. Performance comparisons of different RL algorithms: (a) sum of log rates; (b) the number of users exceeding the user rate threshold.

mobility follows the random direction mobility model [29]. The speed and angle of user mobility follow uniform distributions in $[0, 8\text{m/s}]$ and $[0, 2\pi)$, respectively. The simulation parameters for networks are described in Table 2. The simulation progresses through 500 episodes, each comprising 500 steps. All networks are composed of three hidden layers which are fully connected layers with 500, 1000, and 500 neurons. The activation function of the third hidden layer in the actor-network is customized to generate the action $[\tilde{\mathbf{Y}} \tilde{\mathbf{Z}}]$. A softmax function is used for each row of the RA matrix $\tilde{\mathbf{Y}}$ for DBS links, and a sigmoid function is used for each column of the RA matrix $\tilde{\mathbf{Z}}$ for IAB node links [7]. Regarding hyperparameters, the soft update rate is $\tau = 0.01$, the replay buffer size is 2000, and the mini-batch size is 64.

Fig. 2 shows the performance of the proposed algorithm and compares it with other RL algorithms. Fig. 2(a) shows various RL algorithm performances, including the proposed one, in terms of the *sum of log rates*. This metric accumulates the logarithm of an individual user's average data rate per episode, over all users. Considering that equation (1a) quantifies users' logarithmic satisfaction based on their data rates (i.e., $\log(\tilde{R}_k)$), the sum of log rates presented in Fig. 2(a) is consistent with (1a). For better visualization, all curves are smoothed by averaging results from 20 independent episodes. Fig. 2(a) evaluates performance improvements of RL algorithms in sub-6GHz. The performance of the proposed algorithm, referred to as *DDPG*, converges well and the effectiveness of the mechanism for collaboratively solving the two subproblems, which is presented in Section 4. It demonstrates better performance compared to the algorithm in which all subchannels are fully allocated and the static allocation remains unchanged even in dynamic environments, referred to as *Full Reuse*. Because, DDPG is trained to allocate resources in the direction of mitigating interference, unlike Full Reuse, which always suffers interference from all other BSs. Furthermore, the performance of DDPG is compared with double deep Q-Network [7] and multi-agent DDPG [30], referred to as *DDQN* and *MADDPG*, respectively. DDQN is a value-based RL algorithm that mitigates Q-value overestimation by using two deep neural networks; one for action selection and the other for evaluation. On the other hand, MADDPG is a policy-based multi-agent RL algorithm that extends DDPG to enable decentralized execution through actor networks with centralized training via critic network. DDPG outperforms DDQN, while its performance is nearly identical to MADDPG. However, the execution time of MADDPG is approximately three times longer, making the advantages of the

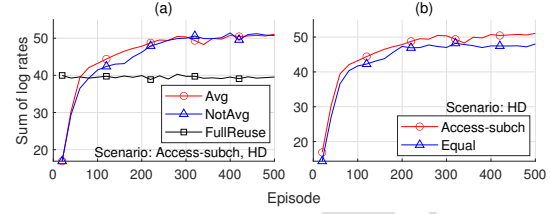


Figure 3. Performance comparisons of proposed methods: (a) the effect of state design; (b) the effect of backhaul resource partition methods.

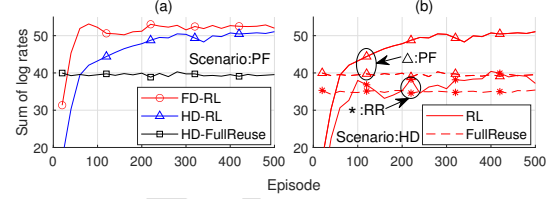


Figure 4. The sum of logarithm of user rates with various scenarios in sub-6GHz: (a) Comparison of FD and HD; (b) Comparison of PF and RR.

multi-agent approach negligible. Fig. 2(b) demonstrates the number of users whose data rate exceeds the given user rate threshold. The user data rate used as the threshold is the average user data rate from the 400th to the 500th episode, and in the case of MADDPG, the user data rates across all agents are averaged. At lower thresholds, DDPG has fewer users satisfying the condition compared to MADDPG. However, as the threshold increases, the number of users satisfying the condition with DDPG surpasses that of MADDPG. Additionally, DDPG outperforms DDQN and Full Reuse across all thresholds, demonstrating its effectiveness in resource allocation.

Fig. 3 evaluates the performance enhancement of the proposed algorithms in sub-6GHz. Fig. 3(a) demonstrates that the proposed state, which averages values (i.e. Avg), performs similarly to NotAvg, where the 'NotAvg' in the legend indicates a state design using raw values without averaging. In other words, Avg significantly reduces the state space without notable degradation in performance. Fig. 3(b) compares two backhaul resource partition methods η_k , which were defined when calculating the data rate of users served by IAB nodes in Section 3, i.e., 'equal partition' referred to as 'Equal', and 'access-subch-based partition' referred to as 'Access-subch'. The results demonstrate that access-subch, which allocates more backhaul resources to users who allocate more access resources, can improve performance than equal. Therefore, unless otherwise stated, using the access-subch is assumed.

Fig. 4 examines the performances depending on duplexing (i.e. full duplex [FD] and half duplex [HD]) and scheduling (i.e. proportional fairness [PF] and round robin [RR]) in sub-6GHz. Fig. 4(a) demonstrates that both FD and HD work well and FD outperforms HD. Fig. 4(b) presents the impact of the scheduling algorithms. Depending on the selected scheduling algorithm, the virtual scheduling works as follows: PF chooses users based on (2), while the RR sequentially chooses users one by one. It is evident that PF outperforms RR and RL dominates

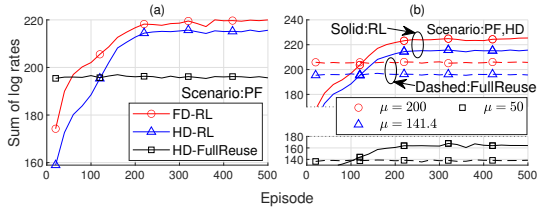


Figure 5. The sum of logarithm of user rates in mmWave: (a) Performance of mmWave; (b) Performance for different μ .

FullReuse. Further, it is quite interesting that there is a synergy between PF and RL. The performance gap between RL and FullReuse when using PF is more significant than when using RR. Similarly, the difference between PF and RR when using RL is larger than when using FullReuse. This positive interaction demonstrates the constructive collaboration of the proposed virtual scheduling and RL-RA.

Fig. 5 evaluates the performance of the proposed algorithm in mmWave bands. Fig. 5(a) depicts that the proposed method still well supports both HD and FD, resulting in significant improvements in the sum of log rates. These results indicate that the proposed algorithm well embraces even mmWave bands, which are characterized by features such as beamforming and harsh propagation losses. Fig. 5(b) exhibits the sum of log rates according to the LoS range constant. Increasing the LoS range may enhance desired signal power while worsening intercell interference. The results demonstrate that the proposed scheme adapts well to various LoS ranges.

6. Conclusion

This paper proposed a reinforcement learning (RL)-based algorithm for joint scheduling and resource allocation (RA) in integrated access and backhaul (IAB) networks. To reduce the complexity of this joint problem, it was decomposed into two subproblems: virtual scheduling and RL-based resource allocation (RL-RA), and they are solved collaboratively. The virtual scheduling could be refined from virtual to actual by the RL-RA, while the RL-RA could overcome the multiuser complexity by the virtual scheduling. In RL-RA, the state representation was carefully designed to reduce the state space. The potential users selected by virtual scheduling are only considered and parameter values are averaged across subchannels without sacrificing performance. Extensive simulations verified the effectiveness of the proposed algorithm across various duplexing modes (i.e., half and full duplex) and frequency bands (e.g., sub-6GHz and mmWave), demonstrating its adaptability to comprehensive network conditions. As future work, it would be interesting to extend the proposed algorithm into more dynamic networks such as unmanned aerial vehicle assisted network.

References

[1] B. Tezergil, E. Onur, Wireless backhaul in 5G and beyond: Issues, challenges and opportunities, *IEEE Commun. Surveys Tuts.* 24 (4) (2022) 2579-2632.

[2] C.-K. Wen, L.-S. Tsai, A. Shojaeifard, P.-K. Liao, K.-K. Wong, C.-B. Chae, Shaping a smarter electromagnetic landscape: IAB, NCR, and RIS in 5G standard and future 6G, *IEEE Commun. Standards Mag.* 8 (1) (2024) 72-78.

[3] C. Madapatha, B. Makki, C. Fang, O. Teyeb, E. Dahlman, M.-S. Alouini, T. Svensson, On integrated access and backhaul networks: Current status and potentials, *IEEE Open J. Commun. Soc.* 1 (2020) 1374-1389.

[4] M. Cudak, A. Ghosh, A. Ghosh, J. Andrews, Integrated access and backhaul: A key enabler for 5G millimeter-wave deployments, *IEEE Commun. Magazine* 59 (4) (2021) 88-94.

[5] C. Saha, M. Afshang, H. S. Dhillon, Bandwidth partitioning and downlink allocation in millimeter wave integrated access and backhaul for 5G, *IEEE Trans. Wireless Commun.* 17 (12) (2018) 8195-8210.

[6] C. Saha, H. S. Dhillon, Millimeter wave integrated access and backhaul in 5G: Performance analysis and design insights, *IEEE J. Sel. Areas Commun.* 37 (12) (2019) 2669-2684.

[7] W. Lei, Y. Ye, M. Xiao, Deep reinforcement learning-based spectrum allocation in integrated access and backhaul networks, *IEEE Trans. Cognit. Commun. Netw.* 6 (3) (2020) 970-979.

[8] P. Bhattacharya, F. Patel, A. Alabdulatif, R. Gupta, S. Tanwar, N. Kumar, R. Sharma, A deep-Q learning scheme for secure spectrum allocation and resource management in 6G environment, *IEEE Trans. Netw. Service Manage.* 19 (4) (2022) 4989-5005.

[9] M. M. Sande, M. C. Hlophe, B. T. Maharaj, Access and radio resource management for IAB networks using deep reinforcement learning, *IEEE Access* 9 (2021) 114218-114234.

[10] Q. Cheng, Z. Wei, J. Yuan, Deep reinforcement learning-based spectrum allocation and power management for IAB networks, in: 2021 IEEE International Conference on Communications Workshops (2021) 1-6.

[11] L. Zhi, N. Hehao, H. Yuanzhi, A. Kang, Z. Xudong, C. Zheng, X. Pei, Self-powered absorptive reconfigurable intelligent surfaces for securing satellite-terrestrial integrated networks, *China Commun.* 21 (9) (2024) 276-291.

[12] Z. Lin, H. Niu, K. An, Y. Wang, G. Zheng, S. Chatzinotas, Y. Hu, Re-fracting RIS aided hybrid satellite-terrestrial relay networks: Joint beamforming design and optimization, *IEEE Trans. Aerosp. Electron. Syst.* 58 (4) (2022) 3717-3724.

[13] Z. Lin, M. Lin, B. Champagne, W.-P. Zhu, N. A.-Dhahir, Secrecy-energy efficient hybrid beamforming for satellite-terrestrial integrated networks, *IEEE Trans. Commun.* 69 (9) (2021) 6345-6360.

[14] Y. He, B. Sheng, H. Yin, D. Yan, Y. Zhang, Multi-objective deep reinforcement learning based time-frequency resource allocation for multi-beam satellite communications, *China Commun.* 19 (1) (2022) 77-91.

[15] M. Chen, A. Liu, N. N. Xiong, H. Song, V. C. M. Leung, SGPL: An intelligent game-based secure collaborative communication scheme for metaverse over 5G and beyond networks, *IEEE J. Sel. Areas Commun.* 42 (3) (2023) 767-782.

[16] S. Mhatre, F. Adelantado, K. Ramantas, C. Verikoukis, Intelligent QoS-aware slice resource allocation with user association parameterization for beyond 5G O-RAN-based architecture using DRL, *IEEE Trans. Veh. Technol.* (2024) 1-4.

[17] R. Liu, R. Yu-Ngok Li, M. Di Renzo, L. Hanzo, A vision and an evolutionary framework for 6G: Scenarios, capabilities and enablers, 2024, arXiv preprint arXiv:2305.13887.

[18] Y. Sadovaya, O. Vikhrova, W. Mao, S.-P. Yeh, O. Semiari, H. Nikopour, Delay-aware link scheduling in IAB networks with dynamic user demands, *IEEE Trans. on Veh. Technol.* 73 (10) (2024) 15125-15139.

[19] F. Meng, P. Chen, L. Wu, J. Cheng, Power allocation in multi-user cellular networks: Deep reinforcement learning approaches, *IEEE Trans. Wireless Commun.* 19 (10) (2020) 6255-6267.

[20] A. Soofinezhadmoghaddam, H. Bakhshi, Joint power allocation and hybrid beamforming of mmWave NOMA systems using DDPG, *IEEE Commun. Lett.* 28 (11) (2024) 2578-2582.

[21] J. Cui, Y. Liu, A. Nallanathan, Multi-agent reinforcement learning-based resource allocation for UAV networks, *IEEE Trans. Wireless Commun.* 19 (2) (2020) 729-743.

[22] X. Chen, Z. Li, W. Ni, X. Wang, S. Zhang, Y. Sun, S. Xu, Q. Pei, Toward dynamic resource allocation and client scheduling in hierarchical federated learning: A two-phase deep reinforcement learning approach, *IEEE Trans. Commun.* 72 (12) (2024) 7798-7813.

[23] W. Liu, R. Wang, T. Zhang, K. Li, W. Li, H. Ishibuchi, X. Liao, Hy-

- bridization of evolutionary algorithm and deep reinforcement learning for multiobjective orienteering optimization, *IEEE Trans. Evol. Comput.* 27 (5) (2023) 1260-1274.
- [24] T. Bai, R. W. Heath, Coverage and rate analysis for millimeter-wave cellular networks, *IEEE Trans. Wireless Commun.* 14 (2) (2015) 1100-1114.
- [25] W. Yu, T. Kwon, C. Shin, Multicell coordination via joint scheduling, beamforming, and power spectrum adaptation, *IEEE Trans. Wireless Commun.* 12 (7) (2013) 1-14.
- [26] N. C. Luong, D. T. Hoang, S. Gong, D. Niyato, P. Wang, Y.-C. Liang, D. I. Kim, Applications of deep reinforcement learning in communications and networking: A survey, *IEEE Commun. Surveys Tuts.* 21 (4) (2019) 3133-3174.
- [27] A. Laterre, Y. Fu, M. K. Jabri, A.-S. Cohen, D. Kas, K. Hajjar, T. S. Dahl, A. Kerkeni, K. Beguir, Ranked reward: Enabling self-play reinforcement learning for combinatorial optimization, 2018, arXiv preprint arXiv:1807.01672.
- [28] E. Chen, M. Tao, N. Zhang, User-centric joint access-backhaul design for full-duplex self-backhauled wireless networks, *IEEE Trans. Commun.* 67 (11) (2019) 7980-7993.
- [29] D. Guo, L. Tang, X. Zhang, Y.-C. Liang, Joint optimization of handover control and power allocation based on multi-agent deep reinforcement learning, *IEEE Trans. Veh. Technol.* 69 (11) (2020) 13124-13138.
- [30] F. Chai, Q. Zhang, H. Yao, X. Xin, F. Wang, M. Xu, Z. Xiong, D. Niyato, Multi-agent DDPG based resource allocation in NOMA-enabled satellite IoT, *IEEE Trans. Commun.* 72 (10) (2024) 6287-6300.

Author Statement

Joeun Kim: Conceptualization, Data curation, Formal analysis, Investigation Methodology, Resources, Software, Validation, Visualization, Writing - original draft, Writing - review & editing

Youngil Jeon: Conceptualization, Investigation Methodology, Project administration, Resources

Junhwan Lee: Funding acquisition, Project administration

Moon-Sik Lee: Funding acquisition, Project administration

Taesoo Kwon: Conceptualization, Formal analysis, Funding acquisition, Investigation Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing - original draft, Writing - review & editing

Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

--